(Im)possibility of Asynchronous MPC with Honest Majority over Blockchains

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Why Asynchronous MPC over Blockchains?

MPC: n parties jointly compute $f(x_1, x_2, \ldots, x_n)$ on private inputs, even when up to t of them are corrupt



Blockchain apps need MPC

Private payments, randomness beacons and threshold wallets all post MPC outputs on-chain



Blockchains are not synchronous

Messages can be delayed arbitrarily, so the MPC running alongside must be asynchronous



Asynchrony costs resilience

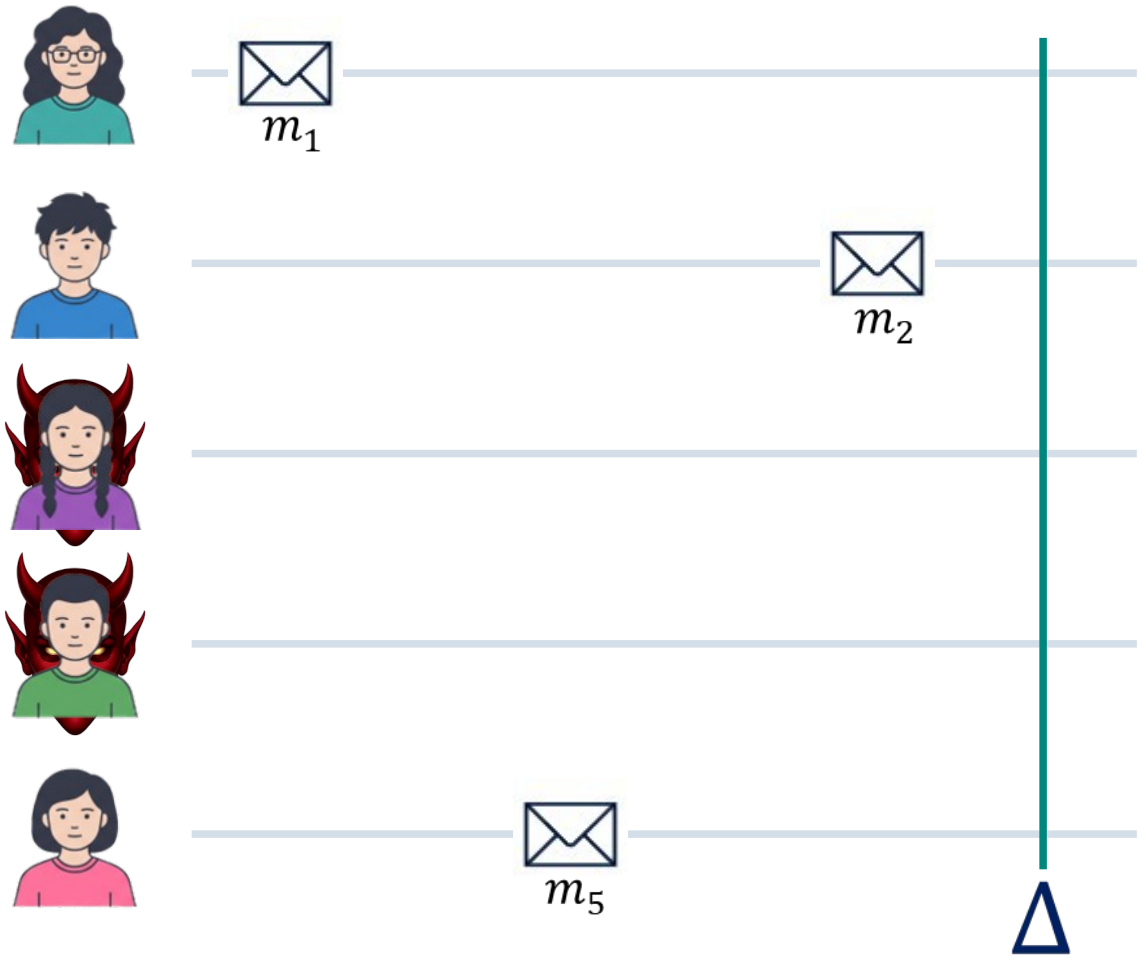
Synchronous MPC: $n > 2t$

Asynchronous MPC: $n > 3t$

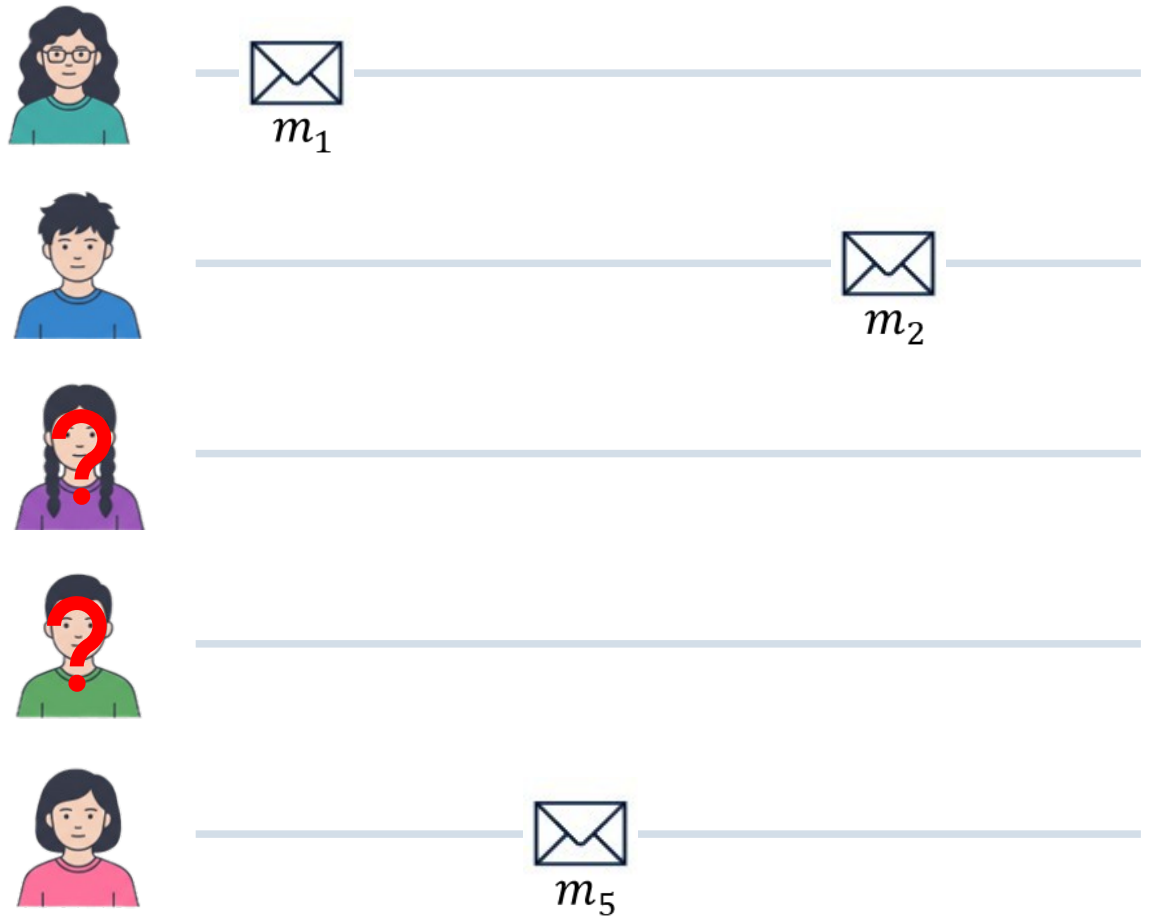
This talk: can access to a blockchain bring AMPC down to an honest majority, $n > 2t$?

Network Setting

Synchronous network



Asynchronous network



Asynchronous Network



m_1 m_5 m_2 m_3 m_4

received not yet received

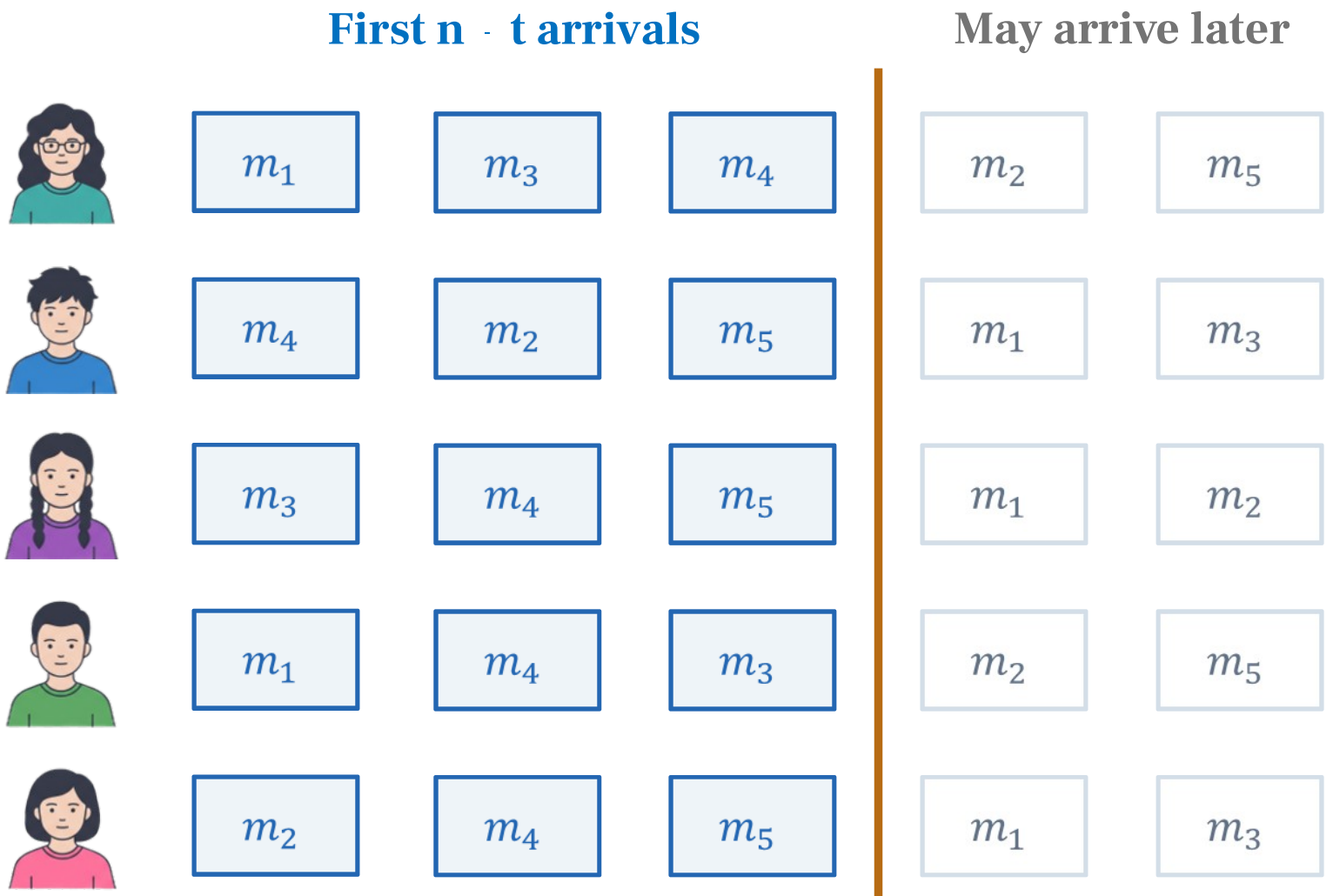
Wait for 5

May never finish

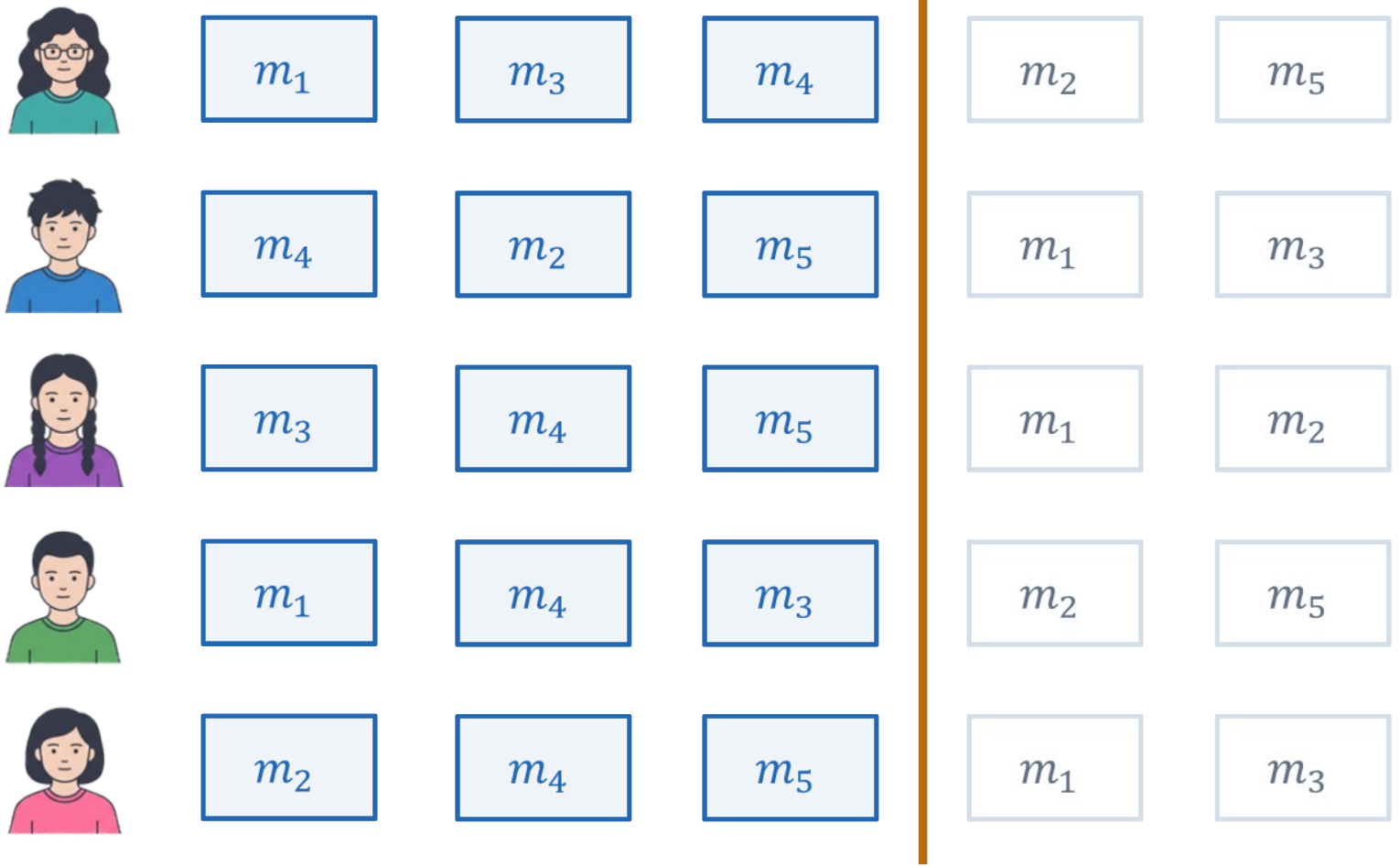
Wait for $3 = n - t$

Enough to keep moving

Agreement on Common Subset (ACS)



Agreement on Common Subset (ACS)



THE PROBLEM

Different parties receives different subsets

THE GOAL

Agree on the same subset of at least $n - t$ contributions

ABA requires $n > 3t$

So AMPC needs a super honest majority, $n > 3t$

unless parties get extra resources

AMPC with Honest Majority

Work	Key Additional Resources
[BHN10]	Limited synchronous broadcast
[BBCK14]	Non-equivocation oracle
[Coh16]	Ideal ABA oracle and TFHE
Our Work	**Blockchains and THE**

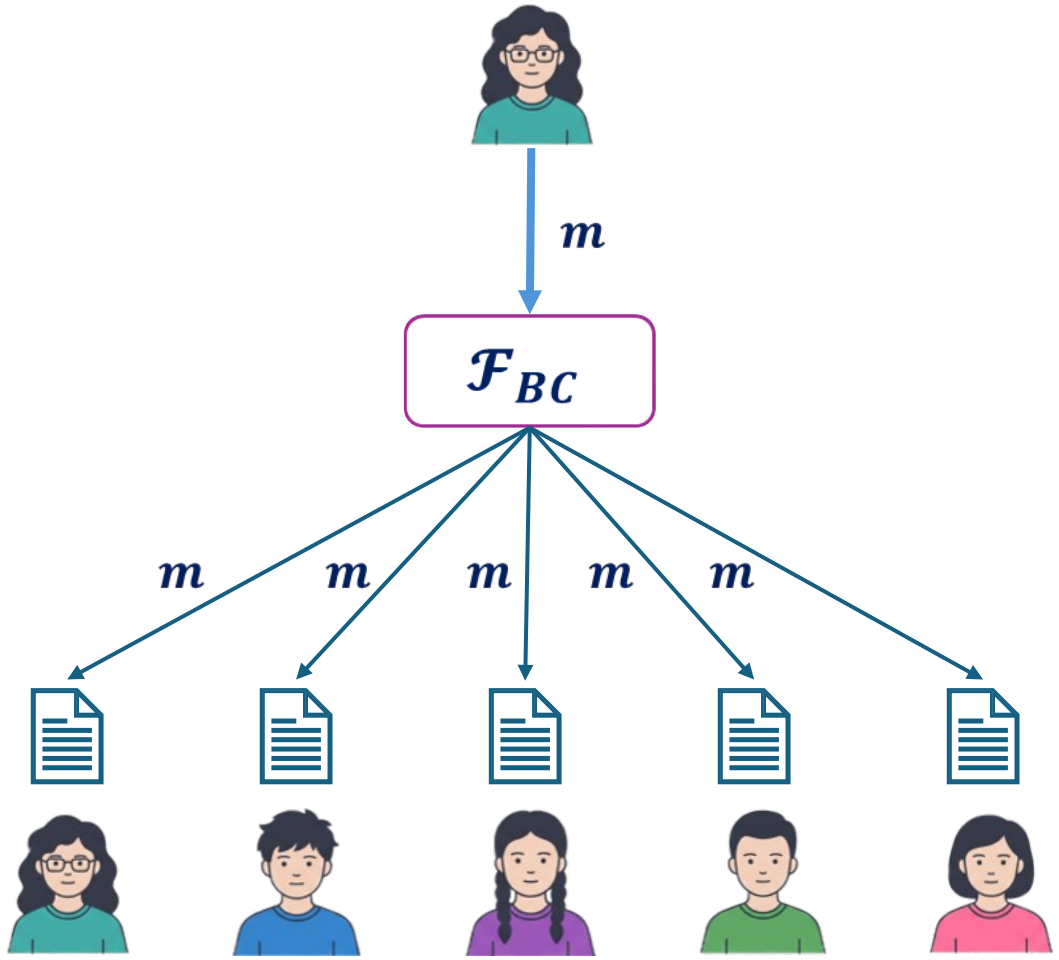
[BHN10]: Zuzana Beerliová-Trubíniová, Martin Hirt, and Jesper Buus Nielsen. On the theoretical gap between synchronous and asynchronous MPC protocols. PODC ’10.

[BBCK14]: Michael Backes, Fabian Bendun, Ashish Choudhury, and Aniket Kate. Asynchronous MPC with a Strict Honest Majority Using Non-equivocation. PODC ’14.

[Coh16]: Ran Cohen. Asynchronous Secure Multiparty Computation in Constant Time. PKC ’16.

Blockchain Functionality

Liveness

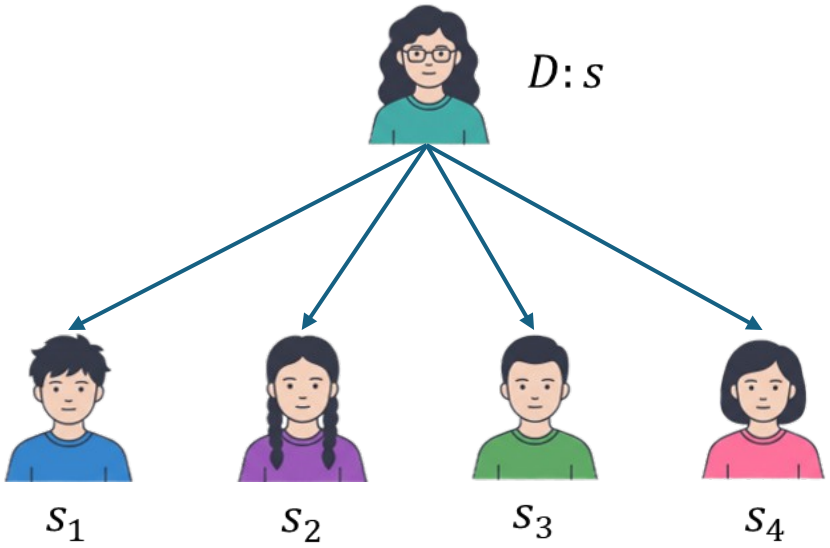


Persistence

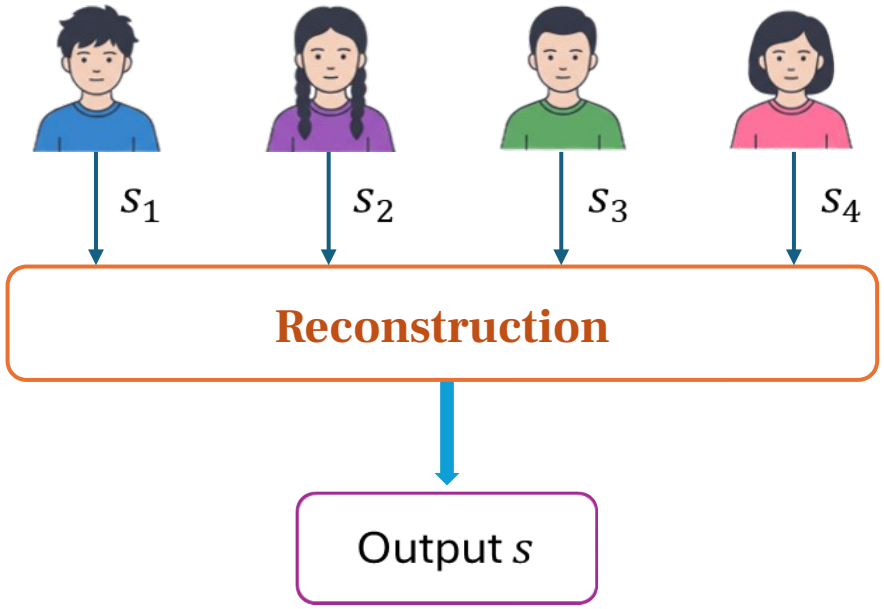


Asynchronous Verifiable Secret Sharing (AVSS)

Sharing Phase



Reconstruction Phase



Properties:

- Termination
- Correctness
- Privacy

Our Results

Cryptographic setting	Impossibility	Possibility
No trusted setup	AVSS against omission adversary for $n \leq 3t$	AVSS against fail-stop adversary with $n = 2t + 1$
Setup ✦ Minicrypt	AMPC even against fail-stop adversary for $n \leq 3t$	AVSS against Byzantine adversary with $n = 2t + 1$
Setup ✦ public key	*AVSS even against fail-stop adversary for $n \leq 2t$	AMPC against Byzantine adversary with $n = 2t + 1$

Corruption types, weakest to strongest



Fail-stop
Follows the protocol, but may crash and stay silent for good



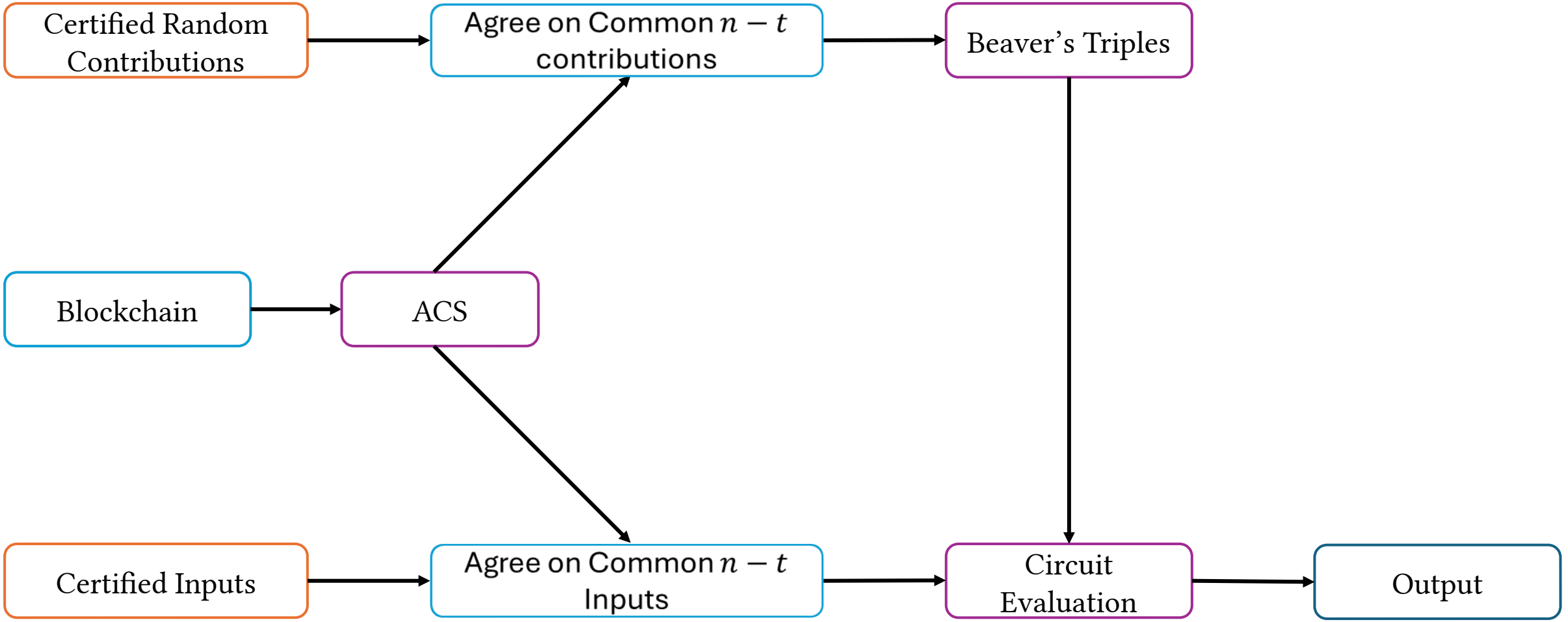
Omission
Follows the protocol, but may drop any message it sends or receives



Byzantine
Deviates from the protocol arbitrarily

*the lower bound of $n \leq 2t$ for AVSS is trivial against the fail-stop adversary

Our Approach



Communication Costs

Primitive / phase	Point-to-point Communication (bits)	On-chain Communication (bits)
ACS	$\mathcal{O}(n^2L + n^3\kappa)$	$\mathcal{O}(n\kappa)$
AMPC preprocessing	$\mathcal{O}(n^2 c_m \kappa)$	$\mathcal{O}(n\kappa)$
AMPC online	$\mathcal{O}\big((nc_m + n^3)\kappa\big)$	$\mathcal{O}(n\kappa)$

L = message length c_m = multiplication gates κ = security parameter

Conclusion

With an ideal asynchronous blockchain

The limit

No trusted setup, or setup + Minicrypt

AMPC remains impossible for $n \leq 3t$

The possibility

Trusted setup + public-key assumptions

Byzantine-secure AMPC for $n > 2t$

The efficiency

$\mathcal{O}(n\kappa)$ **bits on-chain**

Independent of circuit size

Full paper

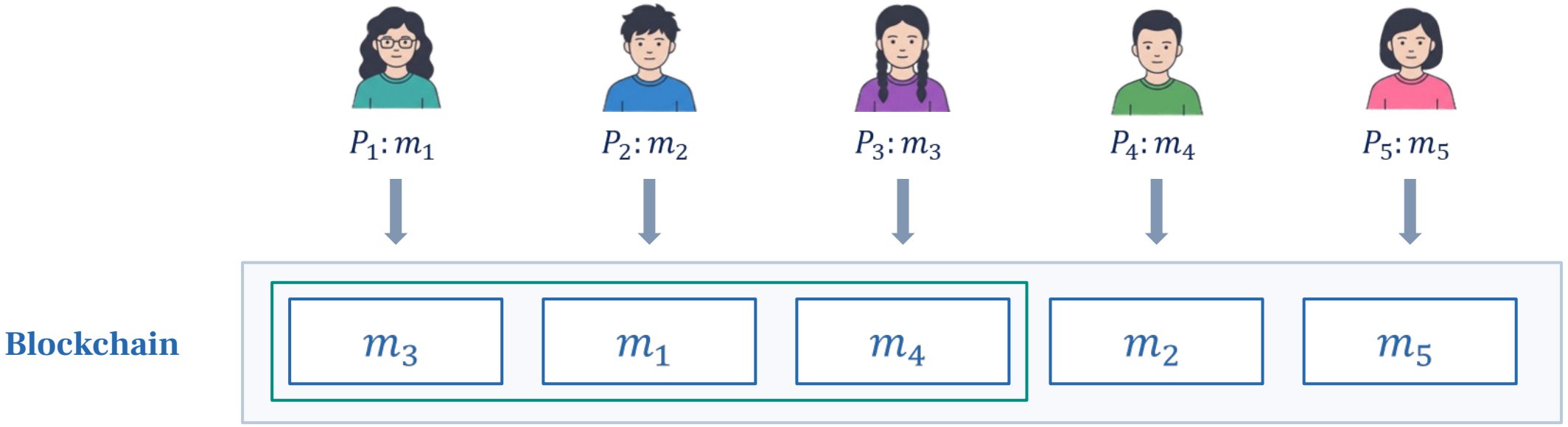


Thank You

Any Questions?

Naïve Approach for ACS

Each party posts its full contribution



Same ledger order; views may advance at different times

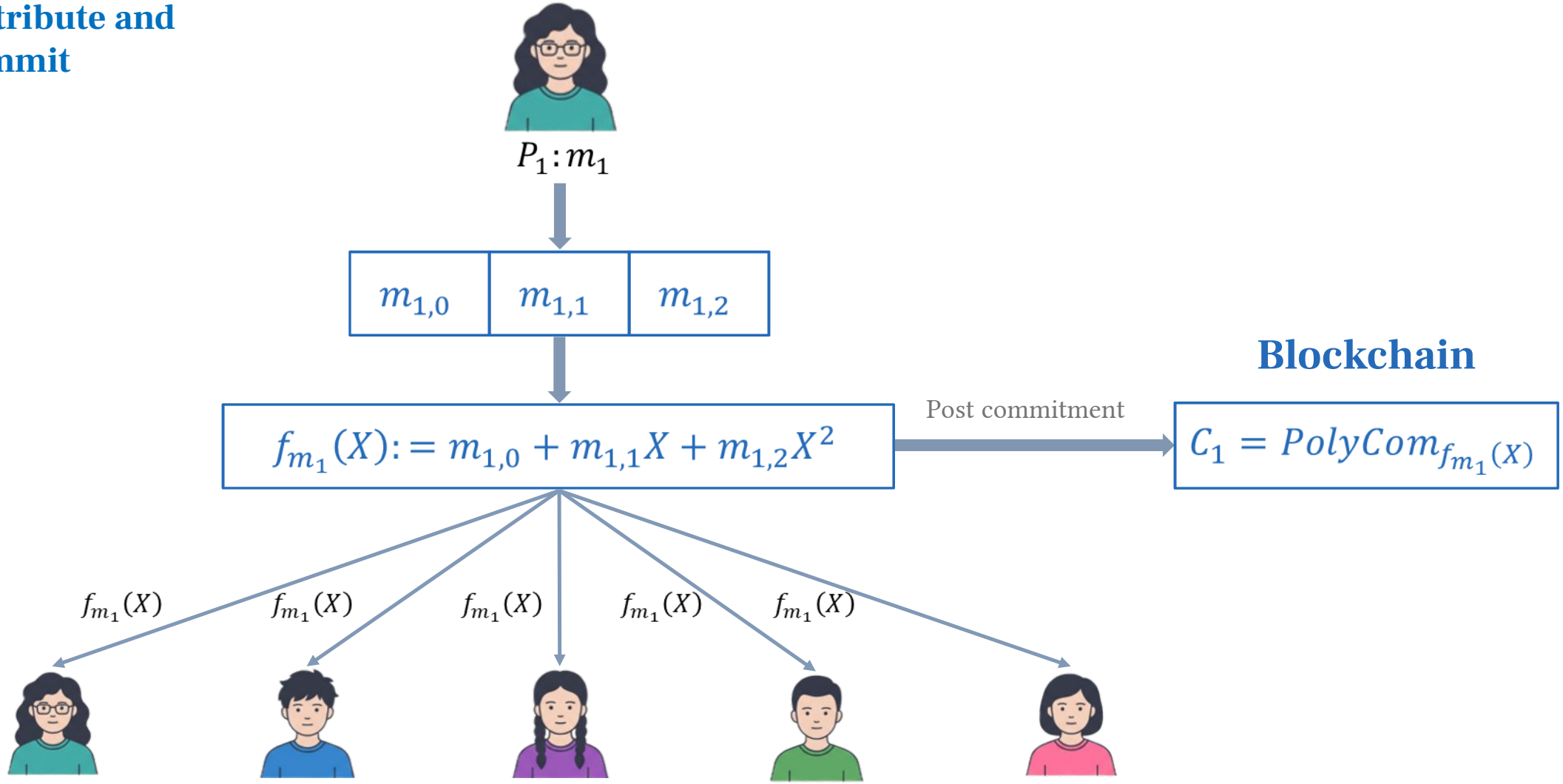
Select the first $n - t$ valid contributions from distinct parties

Every honest party selects $\{m_3, m_1, m_4\}$

L bits per contribution $\Rightarrow$ $\mathcal{O}(nL)$ bits on-chain

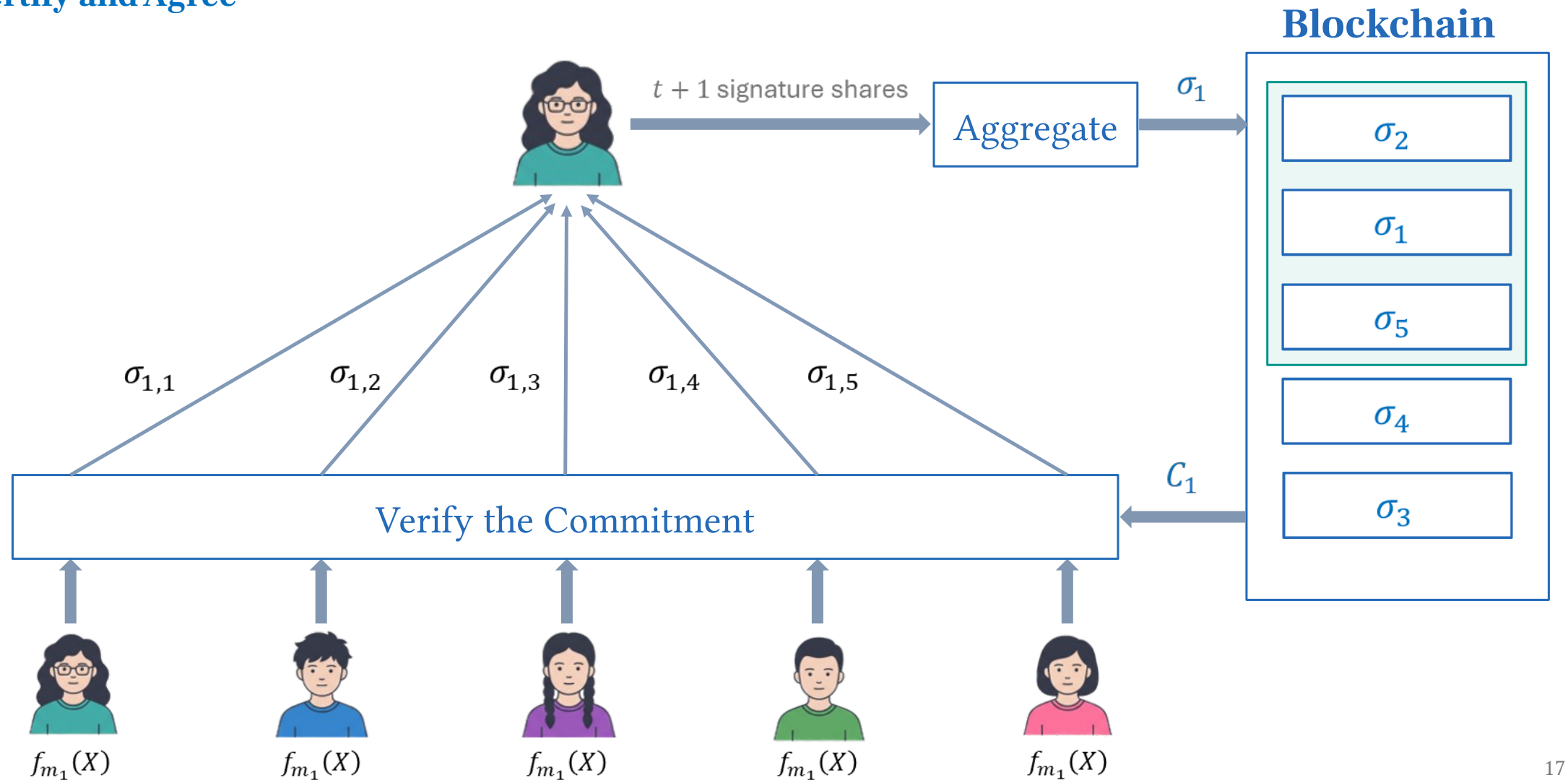
Better Approach for ACS

Distribute and Commit



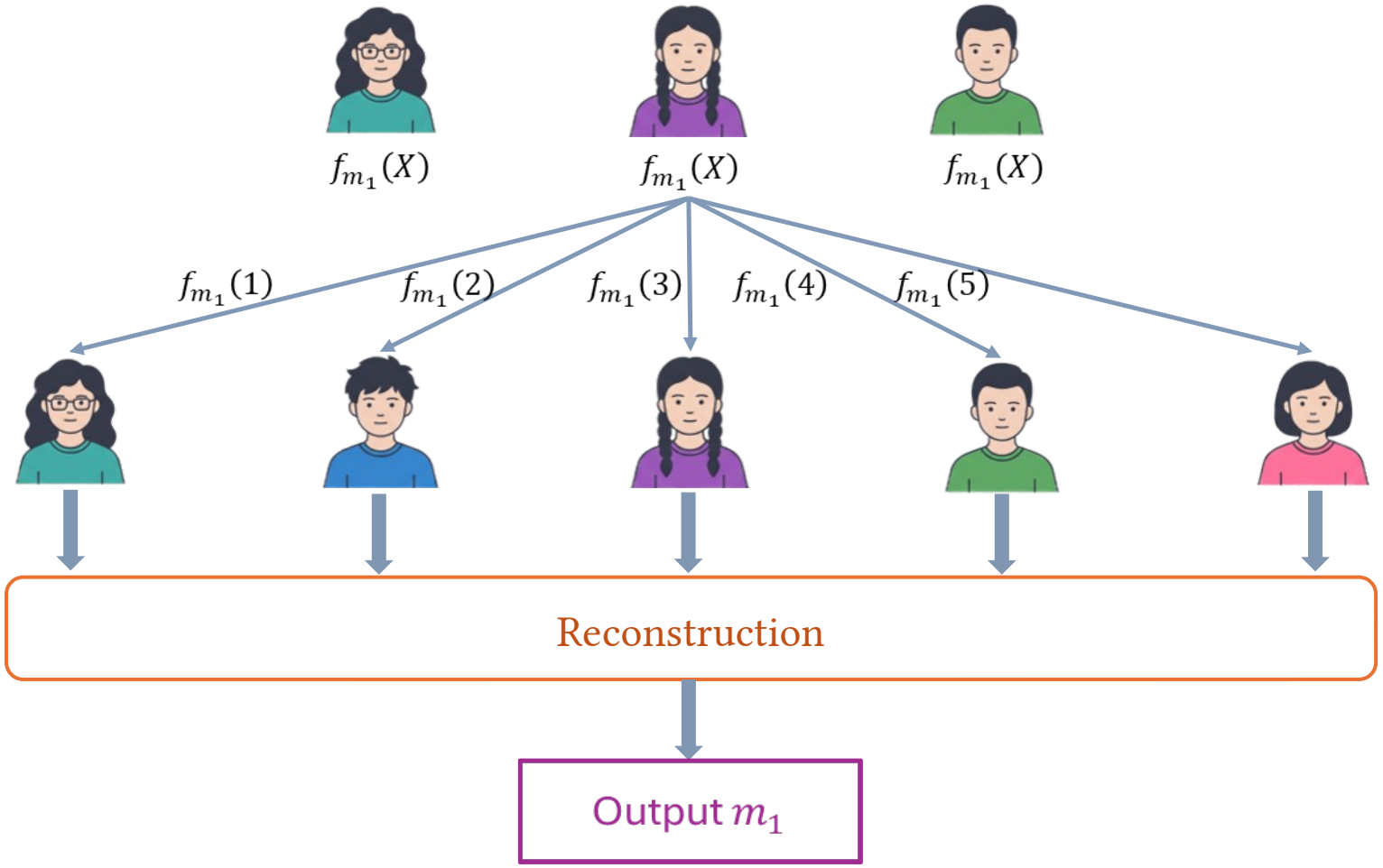
Better Approach for ACS

Certify and Agree



Better Approach for ACS

Relay and Reconstruct

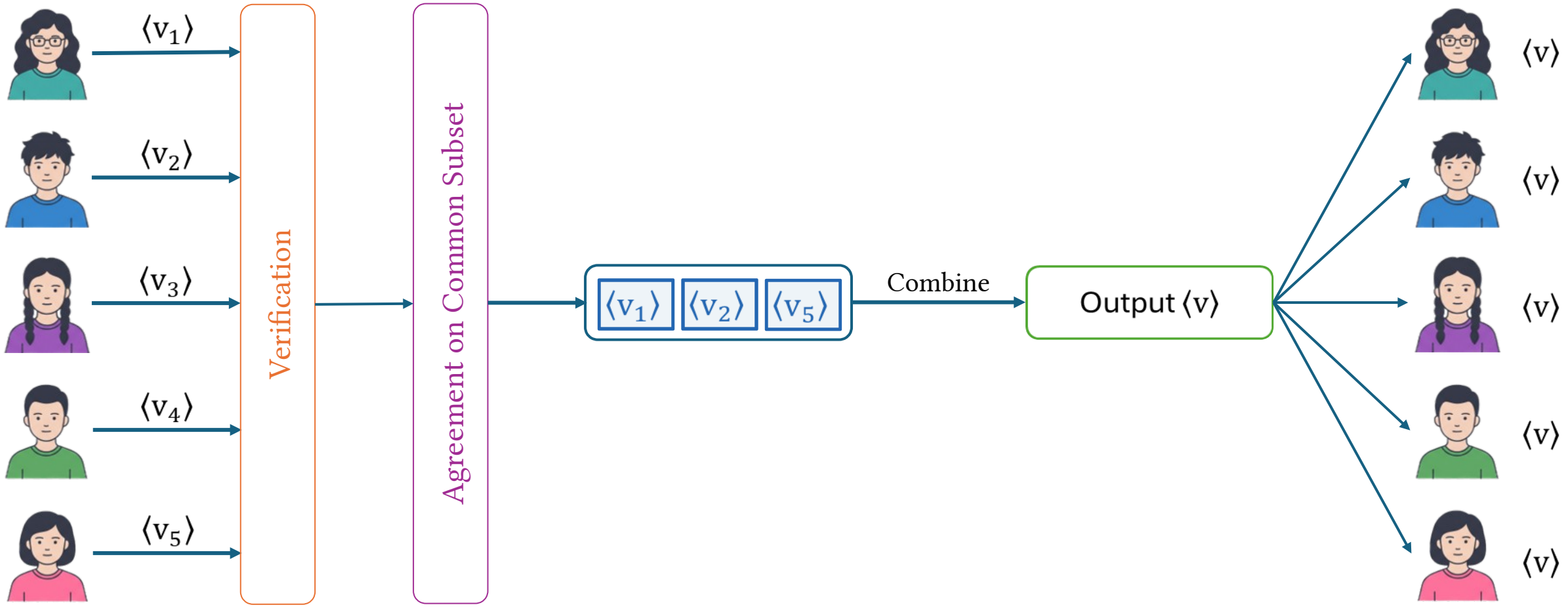


Core Set



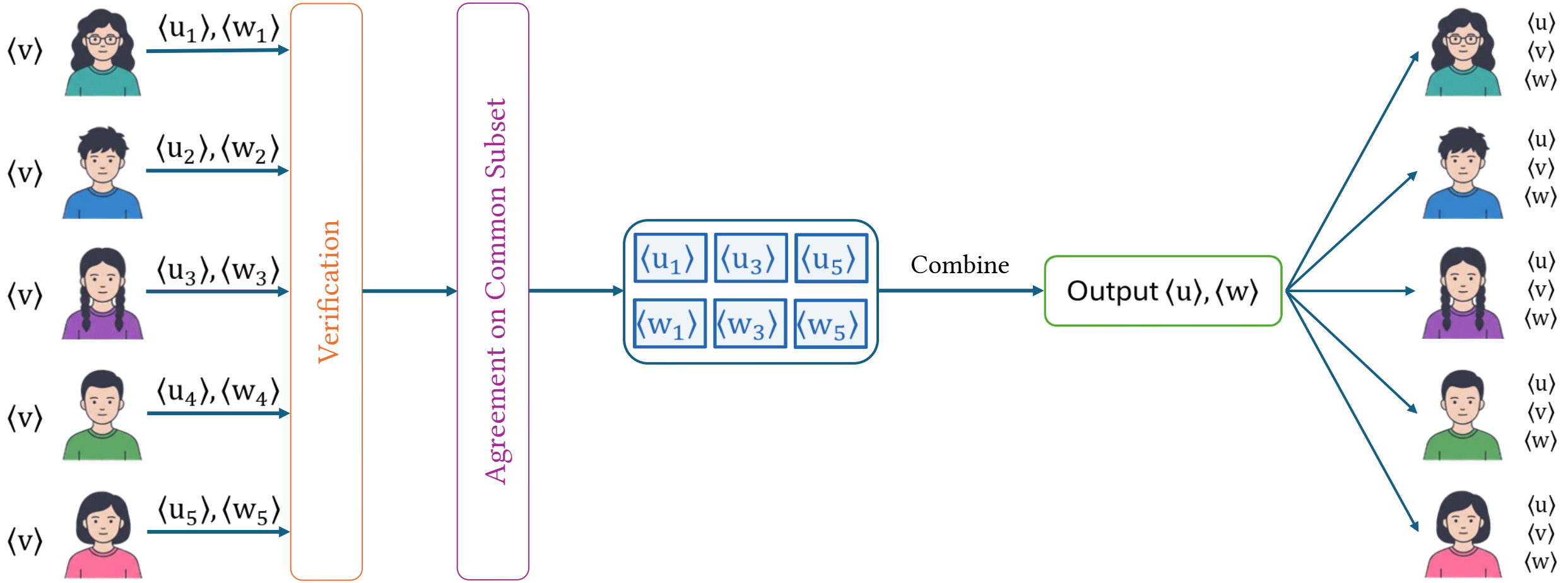
On Chain Communication: $\mathcal{O}(n\kappa)$
κ: security parameter

Pre-processing



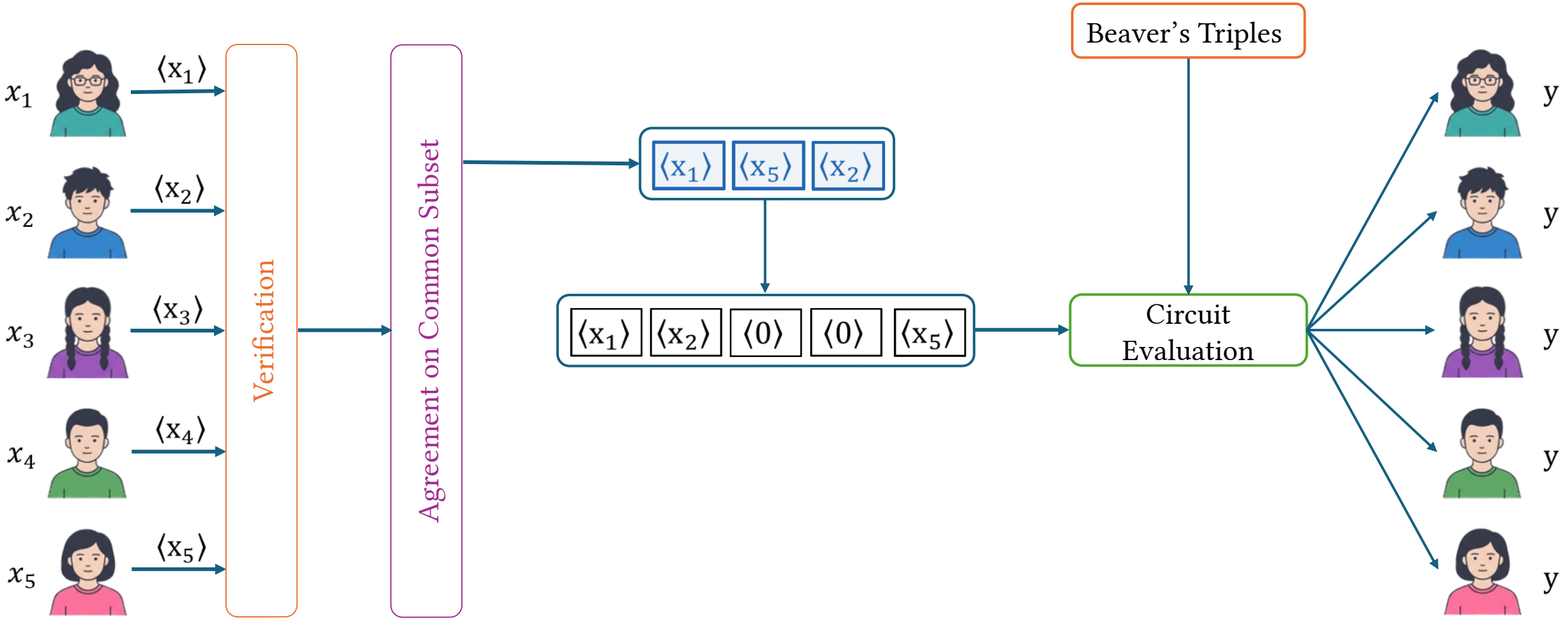
⟨·⟩: Encrypted-and-Committed Values

Pre-processing



⟨·⟩: Encrypted-and-Committed Values

Circuit Evaluation



$\langle \cdot \rangle$: Encrypted-and-Committed Values